

Evaluating named entity recognition tools for extracting social networks from novels

Niels Dekker¹, Tobias Kuhn¹ and Marieke van Erp²

¹ Department of Computer Science, Vrije Universiteit Amsterdam, Amsterdam, The Netherlands

² DHLab, KNAW Humanities Cluster, Amsterdam, The Netherlands

ABSTRACT

The analysis of literary works has experienced a surge in computer-assisted processing. To obtain insights into the community structures and social interactions portrayed in novels, the creation of social networks from novels has gained popularity. Many methods rely on identifying named entities and relations for the construction of these networks, but many of these tools are not specifically created for the literary domain. Furthermore, many of the studies on information extraction from literature typically focus on 19th and early 20th century source material. Because of this, it is unclear if these techniques are as suitable to modern-day literature as they are to those older novels. We present a study in which we evaluate natural language processing tools for the automatic extraction of social networks from novels as well as their network structure. We find that there are no significant differences between old and modern novels but that both are subject to a large amount of variance. Furthermore, we identify several issues that complicate named entity recognition in our set of novels and we present methods to remedy these. We see this work as a step in creating more culturally-aware AI systems.

Subjects Computational Linguistics, Digital Libraries, Network Science and Online Social Networks

Keywords Social networks, Named entity recognition, Evaluation, Digital humanities, Classic and modern literature, Cultural AI

INTRODUCTION

The characters and their relations can be seen as the backbone of any story, and explicitly creating and analysing a network from these relationships can provide insights into the community structures and social interactions portrayed in novels ([Moretti, 2013](#)). Quantitative approaches to social network analysis to examine the overall structure of these social ties, are borrowed from modern sociology and have found their way into many other research fields such as computer science, history, and literary studies ([Scott, 2012](#)). [Elson, Dames & McKeown \(2010\)](#), [Lee & Yeung \(2012\)](#), [Agarwal, Kotalwar & Rambow \(2013\)](#), and [Ardanuy & Sporleder \(2014\)](#) have all proposed methods for automatic social network extraction from literary sources. The most commonly used approach for extracting such networks, is to first identify characters in the novel through Named Entity Recognition (NER) and then identifying relationships between the characters through for example measuring how often two or more characters are mentioned in the same sentence or paragraph.

Many studies use off-the-shelf named entity recognisers, which are not necessarily optimised for the literary domain and do not take into account the surrounding cultural

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Corresponding author

Marieke van Erp,
marieke.van.erp@dh.huc.knaw.nl

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¹ We follow (Sainte-Beuve, 1910) here in defining a classic novel not as one written by the ancient Greeks or Romans ('the classics') but to canonical works.

context. Furthermore, to the best of our knowledge, such studies focus on social network extraction from 19th and early 20th century novels (which we refer to as *classic novels*).¹ Typically, these classic novels are obtained from Project Gutenberg (<http://gutenberg.org/>), where such public domain books are available for free. While beneficial for the accessibility and reproducibility of the studies in question, more recent novels may not imitate these classic novels with respect to structure or style. It is therefore possible that classic novels have social networks that have a structure that is very different from more recent literature. They might differ, for example, in their overall number of characters, in the typical number of social ties any given character has, in the presence or absence of densely connected clusters, or in how closely connected any two characters are on average. Moreover, changes along dimensions such as writing style, vocabulary, and sentence length could prove to be either beneficial or detrimental to the performance of natural language processing techniques. This may lead to different results even if the actual network structures remained the same. Vala et al. (2015) did compare 18th and 19th century novels on the number of characters that appear in the story, but found no significant difference between the two. Furthermore, an exploration of extracted networks can also be used to assess the quality of the extracted information and investigate the structure of the expression of social ties in a novel.

Thus far, we have not found any studies that explore how NER tools perform on a diverse corpus of fiction literature. In this study, we evaluate four different tools on a set of classic novels which have been used for network extraction and analyses in prior work, as well as more recent fiction literature (henceforth referred to as *modern novels*). We need such an evaluation to assess the robustness of these tools to variation in language over time (Biber & Finegan, 1989) and across literary genres. Comparing social networks extracted from corpora consisting of classic and modern novels may give us some insights into what characteristics of literary text may aid or hinder automatic social network extraction and provide indications of cultural change.

As previous work (Ardanuy & Sporleder, 2014) has included works from different genres, in this work we decided to focus on the fantasy/science fiction domain to smooth potential genre differences in our modern books. In our evaluation, we devote extra attention to the comparison between classic and modern fantasy/science fiction in our corpus.

We define the following research questions:

- *To what extent are off-the-shelf NER tools suitable for identifying fictional characters in novels?*
- *Which differences or similarities can be discovered between social networks extracted for different novels?*

To answer our first research question, we first evaluate four named entity recognisers on 20 classic and 20 modern fantasy/science fiction novels. In each of these novels, the first chapter is manually annotated with named entities and coreference relations. The named entity recognisers we evaluate are: (1) BookNLP (Bamman, Underwood & Smith, 2014;

² A gazetteer is a list of names

<https://github.com/dbamman/book-nlp—commit: 81d7a31>) which is specifically tailored to identify and cluster literary characters, and has been used to extract entities from a corpus of 15,099 English novels. At the time of writing, this tool was cited 80 times. (2) Stanford NER version 3.8.0 (*Finkel, Grenager & Manning, 2005*), one of the most popular named entity recognisers in the NLP research community, cited 2,648 times at the time of writing. (3) Illinois Named Entity Tagger version 3.0.23 (*Ratinov & Roth, 2009*), a computationally efficient tagger that uses a combination of machine learning, gazetteers,² and additional features extracted from unlabelled data. At the time of writing, the system was downloaded over 10,000 times. Our last system (4) is IXA-Pipe-NERC version 1.1.1 (*Agerri & Rigau, 2016*), a competitive classifier that employs unlabelled data via clustering and gazetteers that outperformed other state-of-the-art NER tools on their within and out-domain evaluations.

To answer the second research question, we use the recognised named entities to create a co-occurrence network for each novel. Network analysis measures are then employed to compare the extracted networks from the classic and modern novels to investigate whether the networks from the different sets of novels exhibit major differences.

The contributions of this paper are: (1) a comparison and an analysis of four NER on 20 classic and 20 modern novels; (2) a comparison and an analysis of social network analysis measures on networks automatically extracted from 20 classic and 20 modern novels; (3) experiments and recommendations for boosting performance on recognising entities in novels; and (4) an annotated gold standard dataset with entities and coreferences of 20 classic and 20 modern novels.

The remainder of this paper is organised as follows. We first discuss related work in the section ‘Related Work’. Next, we describe our approach and methods in the section ‘Materials and Data Preparation’. We present our evaluation of four different NER systems on 20 classic and 20 modern novels in the section ‘Named Entity Recognition Experiments and Results’, followed by the creation and analysis of social networks in the section ‘Network Analysis’. We discuss issues that we encountered in the identification of fictional characters and showcase some methods to boost performance in the section ‘Discussion and Performance Boosting Options’. We conclude by suggesting directions for future work in the section ‘Conclusion and Future Work’.

The code for all experiments as well as annotated data can be found at <https://github.com/Niels-Dekker/Out-with-the-Old-and-in-with-the-Novel>.

RELATED WORK

As mentioned in the section ‘Introduction’, we have not found any other studies that compared the performances of social network extraction on classic and modern novels; or compared the structures of these networks. This section therefore focuses on the techniques used on classic literature. In first part of this section, we will describe how other studies extract and cluster characters. In the second part, we outline what different choices can be made for the creation of a network, and motivate our choices for this study.

Named entity recognition

The first and foremost challenge in creating a social network of literary characters is identifying the characters. NER is often used to identify passages in text that identify things by a name. Furthermore, identified passages are often also classified into various categories such as *person*, *location*, and *organisation*. Typically, this approach is also used to identify miscellaneous numerical mentions such as dates, times, monetary values, and percentages.

Elson, Dames & McKeown (2010), *Ardanuy & Sporleder (2014)*, *Bamman, Underwood & Smith (2014)*, and *Vala et al. (2015)* all use the Stanford NER tagger (*Finkel, Grenager & Manning, 2005*) to identify characters in literary fiction. On a collection of Sherlock Holmes novels, these studies perform Named entity recognition, F_1 -scores between: 45 and 54. *Vala et al. (2015)* propose that the main difficulty with this collection is the multitude of minor characters, a problem which we expect to be also present in our collections of classic and modern novels.

A big difference between the news domain (for which most language technology tools have been created) and the literary domain, is that names do not have to follow the same ‘rules’ as names in the real world. This topic is explored in the Namescape project by *De Does et al. (2017)* namescape (<http://blog.namescape.nl/>). In this project, one million tokens taken from 550 Dutch novels were manually annotated. A distinction between first and last names was made in order to test whether different name parts are used with different effects. A named entity recogniser was trained specifically for this corpus by namescape-clin, obtaining an F_1 score of 93.60 for persons. The corpus contains fragments of novels written between the 17th and 20th century, but as the corpus and tools are not available, we cannot investigate its depth or compare it directly to our work. Other approaches attempt to use the identification of locations and physical proximity to improve the creation of a social network (*Lee & Yeung, 2012*).

Coreference resolution

One difficulty of character detection is the variety of aliases one character might go by, or: coreference resolution. For example, George Martin’s *Tyrion Lannister*, might alternatively be mentioned as *Ser Tyrion Lannister*, *Lord Tyrion*, *Tyrion*, *The Imp* or *The Halfman*. In the vast majority of cases, it is desirable to collapse those character references into one character entity. However, in some cases, retaining some distinction between character references can be useful: we provide an example of this in subsection ‘Network Exploration’.

Two distinct approaches attempt to address this difficulty, (1) omit parts of a multi-word name, or (2) compile a list of aliases. The former approach leaves out honorifics such as the *Ser* and *Lord* in the above example in order to cluster the names of one character. To automate this clustering step, some work has been done by *Bamman, Underwood & Smith (2014)* and *Ardanuy & Sporleder (2014)*. While useful, the former approach alone provides no solace for the matching of the last two example aliases; where no part of the character’s name is present. The latter approach thus suggests to manually compile a list of aliases for each character with the aid of external resources or annotators. This method is utilised by *Elson, Dames & McKeown (2010)* and

[Lee & Yeung \(2012\)](#). In namespace-clin, wikification (i.e. attempting to match recognised names to Wikipedia resources) is used. Obviously this is most useful for characters that are famous enough to have a Wikipedia page. The authors state in their error analysis [Van Dalen-Oskam et al. \(2014, Section 3.2\)](#) that titles that are most likely from the fantasy domain are most difficult to resolve, which already hints at some differences between names in different genres.

Anaphora resolution

To identify as many character references as possible, it is important to take into account that not all references to a character actually mention the character's name. In fact, [Bamman, Underwood & Smith \(2014\)](#) show that 74% of character references come in the form of a pronouns such as *he*, *him*, *his*, *she*, *her*, and *hers* in a collection of 15,099 English novels. To capture these references, the anaphoric pronoun is typically matched to its antecedent by using the linear word distance between the two, and by matching the gender of anaphora to that of the antecedent. The linear word distance can be, for example, the number of words between the pronoun and the nearest characters. For unusual names, as often found in science fiction and fantasy, identification of the gender may be problematic.

Network creation

For a social network of literary characters, characters are represented by the nodes, whereas the edges indicate to some interaction or relationship. While the definition of a character is uniformly accepted in the literature, the definition of an interaction varies per approach. In previous research, two main approaches can be identified to define such an edge. On the one hand, **conversational networks** are used in approaches by [Chambers & Jurafsky \(2008\)](#), [Elson & McKeown \(2010\)](#), and [He, Barbosa & Kondrak \(2013\)](#). This approach focuses on the identification of speakers and listeners, and connecting each speaker and listener to the quoted piece of dialogue they utter or receive. On the other hand, **co-occurrence networks** (as used by [Ardanuy & Sporleder \(2014\)](#) and [Fernandez, Peterson & Ulmer \(2015\)](#)) are created by connecting characters if they occur in the same body of text. While conversational networks can provide a good view of who speaks directly to whom, [Ardanuy & Sporleder \(2014\)](#) argue that '*...much of the interaction in novels is done off-dialogue through the description of the narrator or indirect interactions*' (p. 34). What value to assign to the edges depends on the end-goal of the study. For example, [Fernandez, Peterson & Ulmer \(2015\)](#) assign a negative or positive sentiment score to the edges between each character-pair in order to ultimately predict the protagonist and antagonist of the text. [Ardanuy & Sporleder \(2014\)](#) used weighted edges to indicate how often two characters interact.

Network analysis

Social network analysis draws upon network theory for its network analysis measures ([Scott, 2012](#)). The application of these measures to networks extracted from literature has been demonstrated insightful in assessing the relationships of characters in for example

‘Alice in Wonderland’ (Agarwal et al., 2012) and ‘Beowulf’, the ‘Iliad’ and ‘Táin Bó Cuailnge’ (‘The Cattle Raid of Cooley’, an Irish epic) (Mac Carron & Kenna, 2012). Network analysis can also play a role in authorship attribution (Amancio, 2015, Akimushkin, Amancio & Oliveira, 2017) and characterising a novel (Elson, Dames & McKeown, 2010).

MATERIALS AND DATA PREPARATION

For the study presented here, we are interested in the recognition and identification of persons mentioned in classic and modern novels for the construction of the social network of these fictitious characters. We use off-the-shelf state-of-the-art entity recognition tools in an automatic pipeline without manually created alias lists or similar techniques. For the network construction, we follow Ardanuy & Sporleder (2014) and apply their co-occurrence approach for the generation of the social network links with weighted edges that indicate how often two characters are mentioned together. We leave the consideration of negative weights and sentiments for future work. Before we will explain the details of the used entity recognition tools, how they compare for the given task, and how their results can be used to build and analyse the respective social networks, we explain first the details of our selected corpus, how we preprocessed the data, and how we collected the annotations for the evaluation.

Corpus selection

Our dataset consists of 40 novels—20 classic and 20 modern novels—the specifics of which are presented in Table A2 in the Appendix. Any selection of sources is bound to be unrepresentative in terms of some characteristics but we have attempted to balance breadth and depth in our dataset. Furthermore, we have based ourselves on selections made by other researchers for the classics and compilations by others for the modern books.

For the classic set, the selection was based on Guardian’s Top 100 all-time classic novels (McCrum, 2003). Wherever possible, we selected books that were (1) analysed in related work (as mentioned in the subsection ‘Coreference Resolution’) and (2) available through Project Gutenberg (<https://www.gutenberg.org/>).

For the modern set, the books were selected by reference to a list compiled by BestFantasyBooksCom (<http://bestfantasybooks.com/top25-fantasy-books.php>, last retrieved: 30 October 2017). For our final selection of these novels, we deliberately made some adjustments to get a wider selection. That is, some of the books in this list are part of a series. If we were to include all the books of the upvoted series, our list would consist of only four different series. We therefore chose to include only the first book of each of such series. As the newer books are unavailable on Gutenberg, these were purchased online. These digital texts are generally provided in .epub or .mobi format. In order to reliably convert these files into plain text format, we used Calibre (<https://calibre-ebook.com/—version 2.78>), a free and open-source e-book conversion tool. This conversion was mostly without any hurdles, but some issues were encountered in terms of encoding, as is discussed in the next section. Due to copyright restrictions, we cannot share this full dataset but our gold standard

Table 1 Annotation example.

Id	Preceding context	Focus sentence	Subsequent context	#	Person 1	Person 2
541	Bran reached out hesitantly	‘Go on’, Robb told him	‘You can touch him’	2	Robb Stark	Bran Stark

annotations of the first chapter of each are provided on this project’s GitHub page. The ISBN numbers of the editions used in our study can be found in [Table A2](#) the Appendix.

Data preprocessing

To ensure that all the harvested text files were ready for processing, we firstly ensured that the encoding for all the documents was the same, in order to avoid issues down the line. In addition, all information that is not directly relevant to the story of the novel was stripped. Even while peripheral information in some books—such as appendices or glossaries—can provide useful information about character relationships, we decided to focus on the story content and thus discard this information. Where applicable, the following peripheral information was manually removed: (1) reviews by fellow writers, (2) dedications or acknowledgements, (3) publishing information, (4) table of contents, (5) chapter headings and page numbers, and (6) appendices and/or glossaries.

During this clean-up phase, we encountered some encoding issues that came with the conversion to plain text files. Especially in the modern novels, some novels used inconsistent or odd quotation marks. This issue was addressed by replacing the inconsistent quotation marks with neutral quotations that are identical in form, regardless of whether if it is used as opening or closing quotation mark.

Annotation

Because of limitations in time and scope, we only annotated approximately one chapter of each novel. In this subsection, we describe the annotation process.

Annotation data

To evaluate the performance for each novel, a gold standard was created manually. Two annotators (not the authors of this article) were asked to evaluate 10 books from each category. For each document, approximately one chapter was annotated with entity co-occurrences. Because the length of the first chapter fluctuated between 84 and 1,442 sentences, we selected an average of 300 sentences for each book that was close to a chapter-boundary. For example, for *Alice in Wonderland*, the third chapter ended on the 315th sentence, so the first three chapters were extracted for annotation. While not perfect, we attempted to strike a balance between comparable annotation lengths for each book, without cutting off mid-chapter.

Annotation instructions

For each document, the annotators were asked to annotate each sentence for the occurrence of characters. That is, for each sentence, identify all the characters in it. To describe this process, an example containing a single sentence from *A Game of Thrones* is included in [Table 1](#). The **id** of the sentence is later used to match the annotated sentence to

Table 2 Annotation instructions.

Guideline	Example
Ignore generic pronouns	‘Everyone knows; you don’t mess with me !’
Ignore exclamations	‘For Christ’s sake!’
Ignore generic noun phrases	‘Bilbo didn’t know what to tell the wizard ’
Include non-human named characters	‘His name is Buckbeak , he’s a hippogriff’

Note:

Boldface indicates an entity mention.

its system-generated counterpart for performance evaluation. The **focus sentence** is the sentence that corresponds to this **id**, and is the sentence for which the annotator is supposed to identify all characters. As context, the annotators are provided with the **preceding** and **subsequent** sentences. In this example, the contextual sentences could be used to resolve the ‘*him*’ in the **focus sentence** to ‘*Bran*’. To indicate how many persons are present, the annotators were asked to fill in the corresponding number (#) of people—with a maximum of 10 characters per sentence. Depending on this number of people identified, subsequent fields became available to the annotator to fill in the character names.

To speed up the annotation, an initial list of characters was created by applying the BookNLP pipeline to each novel. The annotators were instructed to map the characters in the text to the provided list to the best of their ability. If the annotator assessed that a person appears in a sentence, but is unsure of this character’s identity, the annotators would mark this character as *default*. In addition, the annotators were encouraged to add characters, should they be certain that this character does not appear in the pre-compiled list, but occurs in the text nonetheless. Such characters were given a specific tag to ensure that we could retrieve them later for analysis. Lastly, if the annotator is under the impression that two characters in the list refer to the same person, the annotators were instructed to pick one and stick to that. Lastly, the annotators were provided with the peripheral annotation instructions found in Table 2.

While this identification process did include anaphora resolution of singular pronouns—such as resolving ‘*him*’ to ‘*Bran*’—the annotators were instructed to ignore plural pronoun references. Plural pronoun resolution remains a difficult topic in the creation of social networks, as family members may sometimes be mentioned individually, and sometimes their family as a whole. Identifying group membership, and modelling that in the social network structure is not covered by any of the tools we include in our analysis or the related work referenced in the section ‘Related Work’ and therefore left to future work.

NAMED ENTITY RECOGNITION EXPERIMENTS AND RESULTS

We evaluate the performance of four different NER systems on the annotated novels: BookNLP (Bamman, Underwood & Smith, 2014), Stanford NER (Finkel, Grenager & Manning, 2005), Illinois Tagger (Ratinov & Roth, 2009), and IXA-Pipe-NERC (Agerri & Rigau, 2016). The BookNLP pipeline uses the 2014-01-04 release of Stanford NER tagger (Finkel, Grenager & Manning, 2005) internally with the seven-class ontonotes model.

Table 3 Precision (P), Recall (R), and F_1 -scores of different NER systems on classic novels.

Title	BookNLP			Stanford NER			Illinois NER			IXA-NERC		
	<i>P</i>	<i>R</i>	<i>F₁</i>	<i>P</i>	<i>R</i>	<i>F₁</i>	<i>P</i>	<i>R</i>	<i>F₁</i>	<i>P</i>	<i>R</i>	<i>F₁</i>
1984	92.31	70.59	80.00	89.29	73.53	80.65	93.55	85.29	89.23	93.55	85.29	89.23
A Study in Scarlet⊙	25.00	30.77	27.59	22.22	30.77	25.81	14.29	15.38	14.81	20.00	23.08	21.43
Alice in Wonderland	89.13	55.78	68.62	83.33	57.82	68.27	87.07	87.07	87.07	84.30	69.39	76.12
Brave New World	82.93	60.71	70.00	7.50	5.36	6.25	7.69	5.36	6.32	2.63	1.79	2.13
David Copperfield⊙	29.41	35.71	32.26	54.02	67.14	59.87	58.82	71.43	64.52	14.47	15.71	15.07
Dracula⊙	<i>5.00</i>	<i>20.00</i>	<i>8.00</i>	<i>4.00</i>	20.00	6.67	12.50	60.00	20.69	10.53	40.00	16.67
Emma	86.96	93.02	89.89	25.90	27.91	26.87	26.81	28.68	27.72	30.22	32.56	31.34
Frankenstein⊙	52.00	76.47	61.90	37.93	64.71	47.83	30.77	47.06	37.21	34.62	52.94	41.86
Huckleberry Finn	86.84	98.51	92.31	81.08	89.55	85.11	77.92	89.55	83.33	79.71	82.09	80.88
Dr. Jekyll and Mr. Hyde	86.36	82.61	84.44	18.18	17.39	17.78	21.74	21.74	21.74	13.64	13.04	13.33
Moby Dick⊙	67.65	74.19	70.77	63.89	74.19	68.66	68.42	83.87	75.36	37.84	45.16	41.18
Oliver Twist	85.61	94.44	89.81	36.30	42.06	38.97	44.32	33.62	38.24	34.69	40.48	37.36
Pride and Prejudice	79.26	94.69	86.29	32.33	38.05	34.96	29.37	32.74	30.96	33.87	37.17	35.44
The Call of the Wild	80.65	30.49	44.25	86.36	46.34	60.32	89.47	82.93	86.08	88.14	63.41	73.76
The Count of Monte Cristo	78.22	89.77	83.60	67.95	60.23	63.86	79.80	89.77	84.49	72.31	53.41	61.44
The Fellowship of the Ring	73.39	72.15	72.77	66.12	68.35	67.22	56.52	38.40	45.73	63.33	56.12	59.51
The Three Musketeers	65.71	29.49	40.71	63.64	35.90	45.90	45.45	25.64	32.12	73.68	35.90	48.28
The Way We Live Now	73.33	92.77	81.91	49.52	62.65	55.32	28.18	37.35	32.12	43.30	50.60	46.67
Ulysses	76.74	94.29	84.62	70.10	97.14	81.44	71.28	95.71	81.71	72.29	85.71	78.43
Vanity Fair	67.30	65.44	66.36	32.46	34.10	33.26	32.61	34.56	33.56	53.12	47.00	49.88
Mean μ	70.16	68.95	67.72	52.03	53.00	51.13	51.37	55.98	52.26	49.26	48.29	47.61
Standard Deviation σ	24.03	26.27	24.25	27.27	25.24	24.93	28.68	30.16	29.17	29.70	24.71	26.50

Note:

The highest scores in each column are highlighted in **bold**, and the lowest scores in *italics*. Novels written in 1st person are marked with ⊙.

As there have been several releases, and we focus on entities of type Person, we also evaluate the 2017-06-09 Stanford NER four-class CoNLL model.

The results of the different NER systems are presented in Table 3 for the classic novels, and Table 4 for the modern novels. All results are computed using the evaluation script used in the CoNLL 2002 and 2003 NER campaigns using the phrase-based evaluation setup (<https://www.clips.uantwerpen.be/conll2002/ner/bin/conlleval.txt>, last retrieved: 30 October 2017). The systems are evaluated according to micro-averaged precision, recall and F_1 measure. Precision is the percentage of named entities found by the system that were correct. Recall is the percentage of named entities present in the text that are retrieved by the system. The F_1 measure is the harmonic mean of the precision and recall scores. In a phrase-based evaluation setup, the system only scores a point if the complete entity is correctly identified, thus if in a named entity consisting of multiple tokens only two out of three tokens are correctly identified, the system does not obtain any points.

The BookNLP and IXA-Pipe-NERC systems require that part of speech tagging is performed prior to NER, we use the modules included in the respective systems

Table 4 Precision (P), Recall (R), and F_1 scores of different NER systems on modern novels.

Title	BookNLP			Stanford NER			Illinois NER			IXA-NERC		
	<i>P</i>	<i>R</i>	<i>F₁</i>	<i>P</i>	<i>R</i>	<i>F₁</i>	<i>P</i>	<i>R</i>	<i>F₁</i>	<i>P</i>	<i>R</i>	<i>F₁</i>
A Game of Thrones	97.98	62.99	76.68	92.73	66.23	77.27	93.51	93.51	93.51	92.08	60.39	72.94
Assassin's Apprentice⊙	63.33	38.38	47.80	61.19	41.41	49.90	61.45	40.40	48.78	53.12	34.34	41.72
Elantris	82.00	89.78	85.71	76.97	92.70	84.11	83.12	97.08	89.56	76.52	64.23	69.84
Gardens of the Moon	35.29	34.29	34.78	39.02	45.71	42.11	<i>40.43</i>	54.29	46.34	44.44	45.71	45.07
Harry Potter	83.80	90.36	86.96	61.24	65.66	63.37	58.43	58.43	58.43	54.94	53.61	54.27
Magician	72.92	42.17	53.44	65.57	48.19	55.56	77.67	96.39	86.02	63.10	63.86	63.47
Mistborn	96.46	81.95	88.62	93.22	82.71	87.65	90.07	95.49	92.70	94.05	59.40	72.81
Prince of Thorns	69.23	62.07	65.45	64.29	62.07	63.16	60.00	51.72	55.56	72.73	55.17	62.75
Storm Front⊙	65.00	65.00	65.00	68.42	65.00	66.67	64.71	55.00	59.46	63.16	60.00	61.54
The Black Company⊙	77.27	96.23	85.71	<i>29.41</i>	<i>9.43</i>	<i>14.29</i>	67.39	58.49	62.63	60.87	26.42	36.84
The Black Prism	90.29	90.29	90.29	88.35	88.35	88.35	88.68	91.26	89.95	87.21	72.82	79.37
The Blade Itself	62.50	71.43	66.67	71.43	71.43	71.43	52.63	71.43	60.61	55.56	35.71	43.48
The Colour of Magic	83.33	37.50	51.72	84.00	52.50	64.62	71.43	<i>25.00</i>	<i>37.04</i>	77.78	35.00	48.28
The Gunslinger	64.71	100.00	78.57	64.71	100.00	78.57	61.76	95.45	75.00	59.38	86.36	70.37
The Lies of Locke Lamora	86.16	74.05	79.65	87.58	76.22	81.50	86.79	74.59	80.23	88.19	68.65	77.20
The Name of the Wind	85.88	74.49	79.78	87.36	77.55	82.16	78.82	68.37	73.22	85.92	62.24	72.19
The Painted Man	87.02	71.70	78.62	86.47	72.33	78.77	80.81	87.42	83.99	83.09	71.07	76.61
The Way of Kings	80.72	87.01	83.75	75.82	89.61	82.14	70.10	88.31	78.16	66.67	49.35	56.72
The Wheel of Time	66.67	45.86	54.34	70.93	77.71	74.16	58.05	87.26	69.72	66.67	57.32	61.64
Way of Shadows	53.85	77.78	63.64	48.72	70.37	57.58	45.45	92.59	60.98	42.86	44.44	43.64
Mean μ	75.22	69.67	70.86	70.87	67.76	68.17	69.57	74.12	70.09	69.42	55.30	60.54
Standard Deviation σ	15.34	20.73	15.86	17.53	20.95	18.08	15.12	21.57	16.67	15.63	15.02	13.50

Note:

The highest scores in each column are highlighted in **bold**, and the lowest scores in *italics*. Novels written in 1st person are marked with ⊙.

for this. For Stanford NER and Illinois NE Tagger plain text is offered to the NER systems.

As the standard deviations on the bottom rows of [Tables 3](#) and [4](#) indicate, the results on the different books vary greatly. However, the different NER systems generally do perform similarly on the same novels, indicating that difficulties in recognising named entities in particular books is a characteristic of the novels rather than the systems. An exception is *Brave New World* on which BookNLP performs quite well, but the others underperform. Upon inspection, we find that the annotated chapter of this book contains only five different characters among which 'The Director' which occurs 19 times. This entity is consistently missed by the systems resulting in a high penalty. Furthermore, the 'Mr.' in 'Mr. Foster' (occurring 31 times) is often not recognised as in some NE models titles are excluded. A token-based evaluation of Illinois NE Tagger on this novel for example yields a F_1 -score of 51.91. The same issue is at hand with *Dr. Jekyll and Mr. Hyde* and *Dracula*. Although the main NER module in BookNLP is driven by Stanford NER, we suspect that additional domain adaptations in this package account for this performance difference.

When comparing the F_1 -scores of the 1st person novels to the 3rd person novels in [Tables 3 and 4](#), we find that the 1st person novels perform significantly worse than their 3rd person counterparts, at $p < 0.01$. These findings are in line with the findings of [Elson, Dames & McKeown \(2010\)](#).

In the section ‘Discussion and Performance Boosting Options’, we delve further into particular difficulties that fiction presents NER with and showcase solutions that do not require retraining the entity models.

As the BookNLP pipeline in the majority of the cases outperforms the other systems and includes coreference resolution and character clustering, we further utilise this system to create our networks. The results of the BookNLP pipeline including the coreference and clustering are presented in [Table A4](#). One of the main differences in that table is that if popular entities are not recognised by the system they are penalised heavier because the coreferent mentions are also not recognised and linked to the correct entities. This results in scores that are generally somewhat lower, but the task that is measured is also more complex.

NETWORK ANALYSIS

In this section, we explain how the networks were created using the recognised named entities (subsection ‘Network Construction’), followed by an explanation of network analysis measures that we applied to compare the networks (subsection ‘Network Features’). We discuss the results of the analysis (subsection ‘Results of Network Analysis’), as well as present an exploration of the network of one novel in particular to illustrate how a visualisation of a network can highlight particular characteristics of the interactions in the selected novel (subsection ‘Network Exploration’).

Network construction

As explained in the section ‘Related Work’, we opt for the co-occurrence rather than the conversational method for finding the edges of our networks. The body of text that is used to define a co-occurrence differs per approach. Whereas [Fernandez, Peterson & Ulmer \(2015\)](#) define such a relation if characters are mentioned in the same sentence, [Arduany & Sporleder \(2014\)](#) use a paragraph for the same definition. We consider the delineation of what constitutes a paragraph to be too vague for the purpose of this study. While paragraphs are arguably better at conveying who interacts with whom, simply because of their increased length, it also brings forth an extra complexity in terms of their definition. Traditionally, paragraphs would be separated from another by means of a newline followed by an indented first line of the next paragraph. While this format holds for a part of our collection, it is not uniform. Other paragraph formats simply add vertical white space, or depend solely on the content ([Bringinghurst, 2004](#)). Especially because the text files in our approach originate from different online sources—each with their own accepted format—we decided that the added ambiguity should be avoided. For this study, we therefore define that a co-occurrence relationship between two characters exists if they are mentioned in the same sentence. For a co-occurrence of more than two characters, we follow [Elson, Dames & McKeown \(2010\)](#).

That is, a multi-way co-occurrence between four characters is broken down into six bilateral co-occurrences.

For the construction of each social network, the co-occurrences are translated to nodes for characters and edges for relationships between the characters. We thus create a static, undirected and weighted graph. For the weight of each edge, we follow [Ardanuy & Sporleder \(2014\)](#). That is, each edge is assigned a weight depending on the number of interactions between two characters. For the construction of the network, we used NetworkX (<https://networkx.github.io/—v1.11>) and Gephi (<https://gephi.org/—v0.9.1>) to visualise the networks. To ground the network analysis to be presented below, we gathered some overall statistics of the network creation process shown in [Table A3](#) on page 23. As mentioned in the subsection ‘Annotation’, if the annotator decided that a character was definitely present, but unable to assert which character, the occurrence was marked as *default*. The fraction of defaults represents what portion of all identified characters was marked with *default*. The fraction of unidentified characters represents the percentage of characters that were not retrieved by the system, but had to be added by the annotators. Next, we present some overall statistics such as sentence length, the average number of persons in a sentence, and the average fraction of sentences that mention a person. Lastly, we kept track of the total number of annotated sentences, the total number of unique characters and character mentions. The only difference that could be identified between classes is the average sentence length, which was significant at $p < 0.01$. The sentences in classic books are significantly longer than in modern novels, suggesting that there is indeed some difference in writing style. However, other than that, none of the other measures differ significantly. This is useful information, as it helps support that the novels used in either class are comparable, despite their age-gap.

Network features

We analyse the following eight network features:

- (1) **Average degree** is the mean degree of all the nodes in the network. The degree of a node is defined as the number of other nodes the node is connected to. If the degree of a node is zero, the node is connected to no other nodes. The degree of a node in a social network is thus a measure of its social ‘activity’ ([Wasserman & Faust, 1994](#)). A high value—for example, in *Ulysses*—indicates that the characters interact with many different other characters. Contrarily, a low value—for example, in *1984*—indicates that the characters only interact with a small number of other characters.
- (2) **Average Weighted Degree** is fairly similar to the average degree, but especially in the sense of social networks, a distinction must be made. It differs in the sense that the weighted degree takes into account the weight of each of the connecting edges. Whereas a character in our social network could have a high degree—indicating a high level of social activity—if the weights of all those connected edges are relatively small, this suggests only superficial contact. Conversely, while the degree of a character could be low—for example, the character is only connected to two other characters—the two edges could have very large weights, indicating a deep social connection

between the characters. [Newman \(2006\)](#) underlines the importance of this distinction in his work on scientific collaborations. To continue the examples of *Ulysses* and *1984*; while their average degrees are vastly different (with *Ulysses* being the highest of its class and *1984* the lowest), their average *weighted* degrees are comparable.

- (3) **Average Path Length** is the mean of all the possible shortest paths between each node in the network; also known as the geodesic distance. If there is no path connecting two nodes, this distance is infinite and the two nodes are part of different graph components (see item 7, Connected Components). The shortest path between two nodes can be found by using Dijkstra's algorithm ([Dijkstra, 1959](#)). The path length is typically an indication of how efficiently information is relayed through the network. A network with a low path length would indicate that the people in the network can reach each other through a relatively small number of steps.
- (4) **Network Diameter** is the longest possible distance between two nodes in the network. It is in essence the longest, shortest path that can be found between any two nodes in the network, and is indicative of the linear size of the network ([Wasserman & Faust, 1994](#)).
- (5) **Graph density** is the fraction of edges compared to the total number of possible edges. It thus indicates how complete the network is, where completeness would constitute all nodes being directly connected by an edge. This is often used in social network analysis to represent how closely the participants of the network are connected ([Scott, 2012](#)).
- (6) **Modularity** is used to represent community structure. The modularity of a network is '*...the number of edges falling within groups minus the expected number in an equivalent network with edges placed at random*' ([Newman, 2006](#)). Newman shows modularity can be used as an optimisation metric to approximate the number of community structures found in the network. To identify the community structures, we used the Louvain algorithm ([Blondel et al., 2008](#)). The identification of community structures in graph is useful, because the nodes in the same community are more likely to have other properties in common ([Danon et al., 2005](#)). It would therefore be interesting to see if differences can be observed between the prevalence of communities between the classic and modern novels.
- (7) **Connected components** are the number of distinct graph compartments. That is, a graph component is a subgraph in which any two vertices are connected to each other by paths, and which is connected to no additional vertices in the supergraph. In other words, it is not possible to traverse from one component to another. In most social communities, one 'giant component' can typically be identified, which contains the majority of all vertices ([Kumar, Novak & Tomkins, 2010](#)). A higher number of connected components would indicate a higher number of isolated communities. This is different from modularity in the sense that components are more strict. If only a single edge goes out from a subgraph to the supergraph, it is no longer considered a separate component. Modularity attempts to identify those communities that are basically 'almost' separate components.
- (8) **Average clustering coefficient** is the mean of all clustering coefficients. The clustering coefficient of a node can perhaps best be described as 'all-my-neighbours-know-each-other'.

Social networks with a high clustering coefficient (and low average path length) may exhibit **small world** (https://en.wikipedia.org/wiki/Smallworld_experiment) properties (*Watts & Strogatz, 1998*). The small world phenomenon was originally described by Stanley Milgram in his perennial work on social networks (*Travers & Milgram, 1967*).

Results of network analysis

To answer our second research question, we compared the network features presented in the subsection ‘Network Features’ for the social networks of the two different sets of novels. [Table A5](#) on page 25 shows the results. The most striking feature of these results is the wide variance across social networks on all these network measures for both the classic and the modern novels. The size of these network ranges from just 10 nodes to networks more than 50 times as large. The network size alone can also explain at least a large part of the differences in graph density, diameter, and average path length, but also average degree and clustering coefficient show wide variation.

While we can observe large variation overall, there is no clear difference between the two classes, that is, between classic and modern novels. None of the evaluated network features differ significantly between these classes. Graph density is the feature that comes closest to being significant ($p = 0.09$), with our classic novels on average exhibiting denser networks than the modern ones.

In order to better interpret these values, and in order to find out whether this variance in network features is by itself a characteristic property of social networks exposed in novels, or whether this is true for social networks in general, we need a point for comparison. For that purpose, we compare our network results to metrics that have been reported for other social network in the literature. [Table 5](#) shows 10 such networks for comparison, including three small networks on karate club members, football players, and email users (*Telesford et al., 2011*), three medium-sized networks of mathematicians, a larger group of email users, and actors (*Boccaletti et al., 2006*), and four large networks of online platforms (*Mislove et al., 2007*).

We can see that social networks reported elsewhere exhibit a wide variation as well, showing (unsurprisingly) an even much wider range for the network size, with the reported online social networks reaching millions of nodes. Our networks from novels are on the lower end of the size range, with the smallest ones being smaller than the smallest network of our comparison set (Karate). This directly explains why the path lengths are also on the lower end of the range, but with a considerable overlap. With respect to the average degree, our novel networks are covered by the range given by these comparison networks, with even the outliers of our dataset being less extreme than the most extreme cases of the comparison networks. The same holds for the clustering coefficient, except for the outlier for a very small network with a clustering coefficient of 0 (Alice in Wonderland). In summary, we can say that social networks from novels appear to be no different than social networks in general in showing a high variation in basically all network features across different networks. While networks differ much individually, there is no significant fundamental difference between classic and modern novels.

Table 5 Comparison to other social networks.

Network	via	Nodes	Average degree	Clustering coefficient	Average path length
Karate	Telesford et al. (2011)	35 [†]	4.46	0.55	2.41 [†]
Football	Telesford et al. (2011)	115	10.66	0.40	2.51
E-mail	Telesford et al. (2011)	1,133	9.62	0.22	3.60
Math1999	Boccaletti et al. (2006)	58,516	5.00	0.15	8.46 [◊]
e-mail	Boccaletti et al. (2006)	59,812	2.88	0.03 [†]	4.95
Actors	Boccaletti et al. (2006)	225,226	61.00 [◊]	0.79 [◊]	3.65
YouTube	Mislove et al. (2007)	1,157,827	1.81	0.14	5.10
Flickr	Mislove et al. (2007)	1,846,198	1.76	0.31	5.67
Orkut	Mislove et al. (2007)	3,072,441	1.50 [†]	0.17	4.25
LiveJournal	Mislove et al. (2007)	5,284,457 [◊]	1.62	0.33	5.88
	<i>maximum</i>	522	15.77	0.81	3.33
Classic novels	<i>mean</i>	106	6.14	0.60	2.49
	<i>minimum</i>	10	1.66	0.00	1.53
	<i>maximum</i>	314	10.50	0.75	4.06
Modern novels	<i>mean</i>	99	5.50	0.56	2.68
	<i>minimum</i>	27	3.00	0.42	2.22

Notes:

The highest scores in each column are highlighted with a ◊ and the lowest scores with a † for the comparison networks.

Network exploration

In addition to the formal analysis above, we show here a more informal exploration of one of the networks in order to give a more intuitive explanation of our results. For that purpose, we selected the largest network of the modern novels, which is *A Game of Thrones*. A visualisation of that network is shown in [Fig. 1](#). We see that it is a quite dense network with many connections (it has the highest average degree of all modern novels; see [Table A5](#)) and a complex structure. Despite this complexity, the relationship between the main characters of this novel can easily be identified from this visualisation, and one can clearly identify social clusters. Such informal visual explorations should then of course be substantiated with formal analyses, that is, by ranking the edges of the network by their weights and by applying a clustering algorithm in the case of the two given examples. As the readers of this novel might have already spotted, *Dany* resides in a completely different part of the world in this novel, which explains her distance from rest of the network. Moreover, in *A Game of Thrones*, this character does not at any point physically interact with any of the characters in the larger cluster. This highlights a caveat of the use of co-occurrence networks over conversational networks. The character *Dany* does not truly interact with the characters of this main cluster, but is rather name-dropped in conversations between characters in that cluster. Her character ‘co-occurs’ with the characters that drop her name and edges are created to represent that.

To stick with the example of *Dany*, we can also identify two seemingly separate characters, *Dany* and *Daenerys Targaryen* in [Fig. 1](#). These names actually refer to the same entity. As mentioned in the section ‘Related Work’, this issue may be addressed by creating

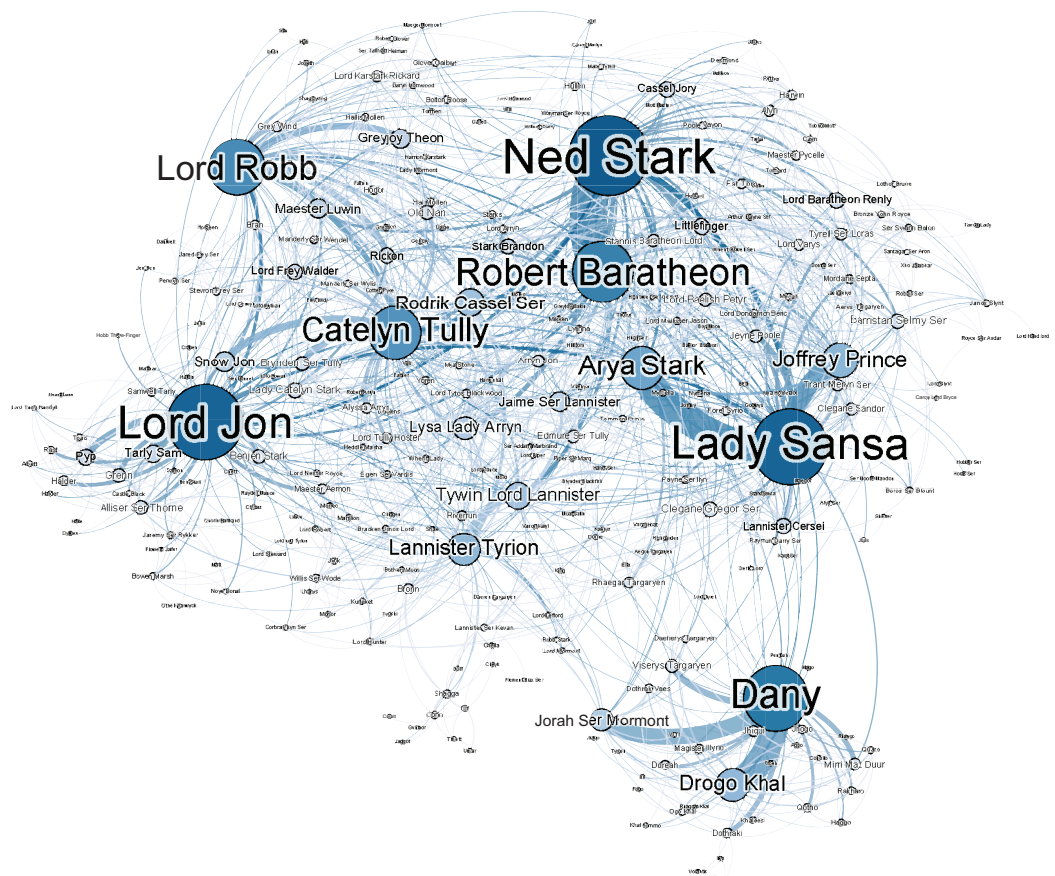


Figure 1 Social network of G.R.R. Martin's *A Game of Thrones*.

Full-size DOI: 10.7717/peerj-cs.189/fig-1

a list of aliases for each character. Some online sources exist that can help expedite this process, but we would argue these sources are not applicable to our modern novels. Whereas 19th century novels typically have characters with more traditional names such as *Elizabeth Bennet*, modern fantasy novels have unconventional names such as *Daenerys Targaryan*. External sources such as on metaCPAN³ can help to connect *Elizabeth* to nicknames such as *Lizzy*, but there are no sources that can do this for *Daenerys* and *Dany*. Even if there was such a source, the question remains whether if it is desirable to collapse those characters. Especially in *A Game of Thrones*, the mentions of *Dany* and *Daenerys Targaryen* occur in entirely different contexts. Whereas references to *Dany* occur in an environment that is largely friendly towards her; her formal name of *Daenerys Targaryen* is mostly used by her enemies (in her absence). Rather than simply collapsing the two characters as one, it might be useful to be able to retain that distinction. This is a design choice that will depend on the type of research question one wants to answer by analysing the social networks.

DISCUSSION AND PERFORMANCE BOOSTING OPTIONS

In analysing the output of the different NER systems, we found that some types of characters were particularly difficult to recognise. Firstly, we found a number of

³ MetaCPAN is a search engine for Perl code and documentation: <https://metacpan.org/source/BRIANL/Lingua-EN-Nickname-1.14/nicknames.txt> (last retrieved: 30 October 2017).

Table 6 Unidentified names in *The Black Company* replaced by generic English names.

Original	Adjusted
Blue	Richard
Croaker	Thomas
Curly	Daniel
Dancing	Edward
Mercy	Charles
One-Eye	Timothy
Silent	James
Walleye	William

unidentified names that are so called word names (i.e. terms that also occur in dictionaries, for example to denote nouns such as *Grace* or *Rebel*). We suspected that this might hinder the NER, which is why we collected all such names in our corpus in Table A1 on page 21, and highlighted such word names with a †. This table shows that approximately 50% of all unidentified names in our entire corpus consist at least partially of a word name, which underpins that this issue is potentially widely spread. In order to verify this, we replaced all potentially problematic names in the source material by generic English names. We made sure not to add names that were already assigned to other characters in the novel, and we ensured that these names were not also regular nouns. An example of these changed character names can be found in Table 6, which shows all names affected for *The Black Company*.

Secondly, we noticed that persons with special characters in their names can prove difficult to retrieve. For example, names such as *d'Artagnan* in *The Three Musketeers* or *Shai'Tan* in *The Wheel of Time* were hard to recognise for the systems. To test this, we replaced all names in our corpus such as *d'Artagnan* or *Shai'Tan* with *Dartagnan* and *Shaitan*. By applying these transformations to our corpus, we found that the performances could be improved, uncovering some of the issues that plague NER. As can be observed in Fig. 2, not all of the novels were affected by these transformations. Out of the 40 novels used in this study, we were able to improve the performance for 14. While the issue of the apostrophed affix was not as recurrent in our corpus as the real-word names, its impact on performance is troublesome nonetheless. Clearly, two novels are more affected by these transformations than the others, namely: *The Black Company* and the *The Three Musketeers*. To further sketch these issues, we delve a bit deeper into these two specific novels.

These name transformations show that the real-word names and names with special characters were indeed problematic and put forth a problem for future studies to tackle. As illustrated by Fig. 2, the aforementioned issues are also present in the classic novels typically used by related works (such as *The Three Musketeers*). This begs the question of the scope of these problems. To the best of our knowledge, similar works have not identified this issue to affect their performances, but we have shown that with a relatively simple workaround, the performance can be drastically improved. It would thus be interesting to evaluate how much these studies suffer from the same issue. Lastly, as

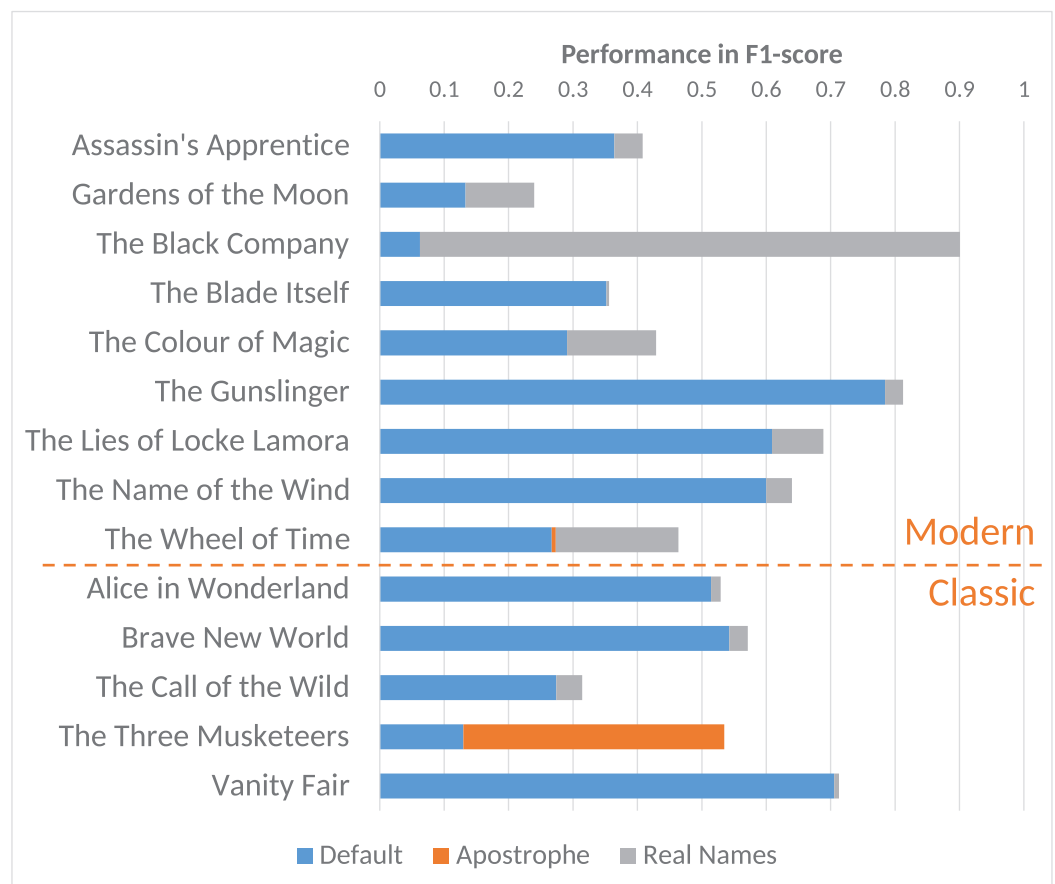


Figure 2 Effect of transformations on all affected classic and modern novels in F_1 -score in using the BookNLP pipeline (includes co-reference resolution). [Full-size !\[\]\(5f471a71b78d7676bc356df190b88ab4_img.jpg\) DOI: 10.7717/peerj-cs.189/fig-2](https://doi.org/10.7717/peerj-cs.189/fig-2)

manually replacing names is clearly far from ideal, we would like to encourage future work to find a more robust approach to resolve this issue.

The Black Company

This fantasy novel describes the dealings of an elite mercenary unit—*The Black Company*—and its members, all of which go by code names such as the ones in [Table 6](#). With a preliminary F_1 -score of 06.85 (see [Table A4](#)), *The Black Company* did not do very well. We found this book had the highest percentage of unidentified characters of our collection. Out of the 14 characters found by our annotators, only five were identified by the pipeline. Interestingly enough, eight out of the nine unidentified characters in this novel have names that correspond to regular nouns. By applying our name transformation alone, the F_1 -score rose from 06.85 to the highest in our collection to 90.00.

The Three Musketeers

This classic piece recounts the adventures of a young man named *d'Artagnan*, after he leaves home to join the Musketeers of the Guard. With an F_1 -score of 13.91 (see [Table A4](#)), *The Three Musketeers* performs the second worst of our corpus, and the worst in its class. By simply replacing names such as *d'Artagnan* with *Dartagnan* the F_1 -score rose

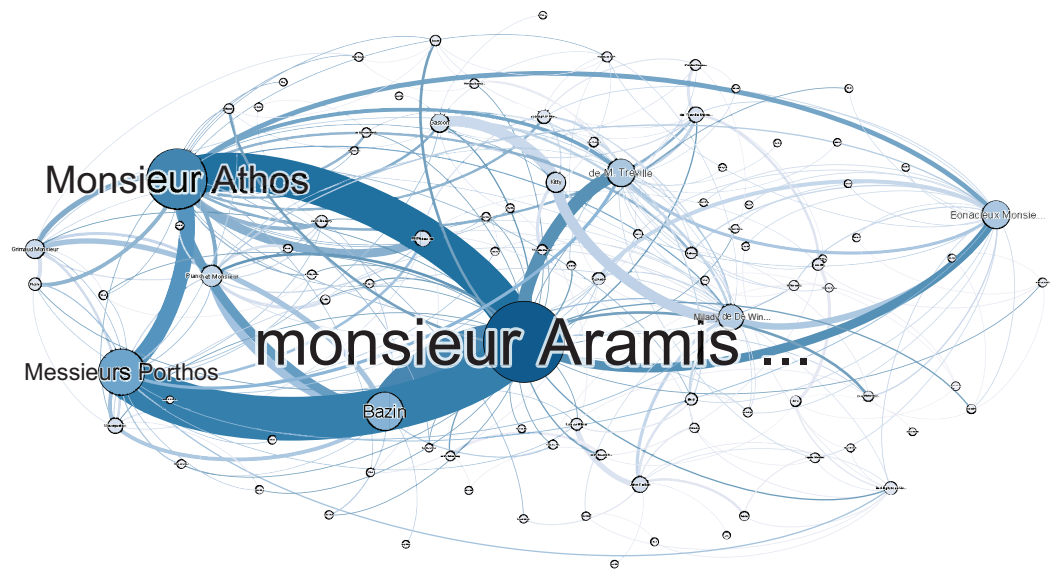


Figure 3 Social network of *The Three Musketeers* without adjustment for apostrophed names.

Full-size [DOI: 10.7717/peerj-cs.189/fig-3](https://doi.org/10.7717/peerj-cs.189/fig-3)

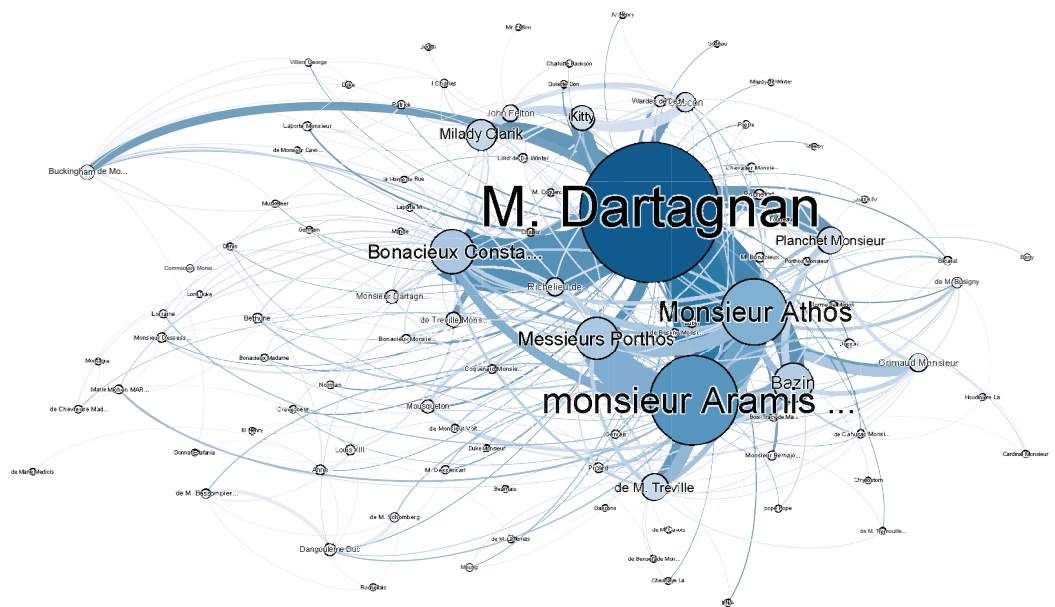


Figure 4 Social network of *The Three Musketeers* with adjustment for apostrophed names.

Full-size [DOI: 10.7717/peerj-cs.189/fig-4](https://doi.org/10.7717/peerj-cs.189/fig-4)

from 13.91 to 53, suggesting that the apostrophed name was indeed the main issue. To visualise this, we have included figures of both *The Three Musketeer* networks—before and after the fix—in Figs. 3 and 4. As can be observed in Fig. 3, the main character of the novel is hardly represented in this network, which is not indicative of the actual story. The importance of resolving the issue of apostrophed named is made clear in Fig. 4, where the main character is properly represented.

CONCLUSION AND FUTURE WORK

In this study, we set out to close a gap in the literature when it comes to the evaluation of NER for the creation of social networks from fiction literature. In our exploration of related work, we found no other studies that attempt to compare networks from classic and modern fiction. To fill this gap, we attempted to answer the following two research questions:

- *To what extent are off-the-shelf NER tools suitable for identifying fictional characters in novels?*
- *Which differences or similarities can be discovered between social networks extracted for different novels?*

To answer our primary research question, we evaluated four state-of-the-art NER systems on 20 classic and 20 modern science fiction/fantasy novels. In our study, we found no significant difference in performance of the named entity recognisers on classic novels and modern novels. We did find that novels written in 3rd person perspective perform significantly better than those written in 1st person, which is in line with findings in related studies. In addition, we observed a large amount of variance within each class, even despite our limitation for the modern novels to the fantasy/science fiction genre. We also identified some recurring problems that hindered NER. We delved deeper into two such problematic novels, and find two main issues that overarch both classes. Firstly, we found that word names such as *Mercy* are more difficult to identify to the systems. We showed that replacing problematic word names by generic placeholders can increase performance on affected novels. Secondly, we found that apostrophed names such as *d'Artagnan* also prove difficult to automatically identify. With fairly simple methods that capture some cultural background knowledge, we circumvented the above two issues to drastically increase the performance of the used pipeline. To the best of our knowledge, none of the related studies discussed in the section 'Related Work' acknowledge the presence of these issues. We would thus like to encourage future work to evaluate the impact of these two issues on existing studies, and call to develop a more robust approach to tackle them in future studies.

To answer our secondary research question, we created social networks for each of the novels in our collection and calculated several networks features with which we compared the two classes. As with the NER experiments, no major differences were found between the classic and modern novels. Again, we found that the distribution of network measures within a class was subject to high variance, which holds for our collection of both classic and modern novels. We therefore recommend that future work focuses on determining particular characteristics that can influence these analyses first and then perform a comparative analysis between subsets to see if this similarity between classes holds when the variance is reduced. Future studies could therefore attempt to compare classic and modern novels in the same genre or narration type (e.g. first-person vs third-person perspective). Lastly, different types of networks that for example collapse characters that occur under different names (cf. Dany and Daenerys) as well as dealing with plural pronouns and group membership (e.g. characters sometimes mentioned

individually and sometimes as part of a group) are currently unsolved problems for language technology and knowledge representation. These issues point to a strong need for more culturally-aware artificial intelligence.

APPENDIX: ADDITIONAL STATISTICS

Table A1 Characters that were not identified by the system, supplied by the annotators.

Classic			Modern	
Ada	Howard	Mrs. Billington	Archmage of Ymitury [†]	Manie
Algy	Joanna	Mrs. Birch [†]	August [†]	Meena
Alice	Johnny	Mrs. Crisp [†]	Bil Baker [†]	Mercy [†]
Anna Boleyn	Jolly Miller [†]	Mrs. Effington Stubbs	Blue [†]	Mrs. Potter [†]
Aprahamian	Leonard	Mrs. Thingummy	Brine Cutter [†]	Old Cob [†]
Belisarius	Lord Mayor [†]	Murray	Bug [†]	One-Eye [†]
Best-Ingram	Lory [†]	Nathan Swain [†]	Chyurda	Pappa Doc [†]
Cain	Major Dover [†]	Peter Teazle [†]	Cotillion [†]	Patience [†]
Caroline	Marie Antoinette	Policar Morrel [†]	Croaker [†]	Plowman [†]
Catherine	Marshal Bertrand [†]	President West [†]	Curly [†]	Poul
Cato	Matilda Carbury	Queequeg	Dadda	Rand [†]
Cervantes	Matron [†]	Rip Van Winkle [†]	Dancing [†]	Shalash
Christine	Miss Birch [†]	Royce	Domi	Shrewd [†]
Chuck Loyola [†]	Miss Crump [†]	Sawbones [†]	Dow [†]	Silent [†]
Cleopatra	Miss Hopkins [†]	Semiramis	Elam Dowtry	Sirius [†]
Connolly Norman [†]	Miss King [†]	Shep	Elaio	Talanel
Curly [†]	Miss Saltire [†]	Sir Carbury	Fredor	Talanelat
Dante	Miss Swindle [†]	Skrimshander [†]	Gart	Ted
Dave	Mme. D'Artagnan	Stamford	Harold	The Empress [†]
Dives [†]	Mollie	Stigand	Harvey	Themos Tresting
Dodo [†]	Mouse [†]	Sudeley	Howard	Theron
Dr. Floss [†]	Mr. Stroll [†]	Swubble	Ien	Threetrees
Duck [†]	Mr. Thursgood	The Director [†]	Ilgrand Lender [†]	Toffston
Edgar Atheling [†]	Mr. Beaufort [†]	Tommy Barnes	Ishar	Verus
Elmo	Mr. Crisp [†]	Unwin	Ishi	Walleye [†]
Farmer Mitchell [†]	Mr. Flowerdew	Ursula	Jim McGuffin [†]	Weasel [†]
Father Joseph [†]	Mr. Lawrence	Victor [†]	Kerible the Enchanter [†]	Willum
Fury [†]	Mr. Morris	Vilkins	Lilly [†]	Wit Congar [†]
Ginny	Mrs Loveday	Von Bischoff		
Henry VIII	Mrs. Bates [†]	Ysabel		
39 out of 90 characters: 43%			30 out of 56 characters: 54%	

Note:

Characters whose names (partly) consist of a real word—such as ‘Curly’ or ‘Mercy’—are marked with a †. Checked against <http://dictionary.com>.

Table A2 Classic and modern novels included in this study.

Classic			
Title	Author	(Year)	E-book No./ISBN
1984	George Orwell	(1949)	9780451518651
A Study in Scarlet	Conan Doyle	(1886)	244
Alice in Wonderland	Lewis Carroll	(1884)	19033
Brave New World	Aldous Huxley	(1865)	9780965185196
David Copperfield	Charles Dickens	(1931)	766
Dracula	Bram Stoker	(1850)	345
Emma	Jane Austen	(1897)	158
Frankenstein	Mary Shelley	(1815)	84
Huckleberry Finn	Mark Twain	(1818)	76
Jekyll and Hyde	Robert Stevenson	(1851)	42
Moby Dick	Herman Melville	(1838)	2701
Oliver Twist	Charles Dickens	(1813)	730
Pride and Prejudice	Jane Austen	(1886)	1342
The Call of the Wild	Jack London	(1903)	215
The Count of Monte Cristo	Alexandre Dumas	(1844)	1184
The Fellowship of the Ring	J. R. R. Tolkien	(1954)	9780547952017
The Three Musketeers	Alexandre Dumas	(1844)	1257
The Way We Live Now	Anthony Trollope	(1875)	5231
Ulysses	James Joyce	(1922)	4300
Vanity Fair	William Thackeray	(1847)	599
Modern			
A Game of Thrones	G.R.R. Martin	(1996)	9780307292094
Assassin's Apprentice	Robin Hobb	(1995)	9781400114344
Elantris	Brandon Sanderson	(2005)	9780765383105
Gardens of the Moon	Steven Erikson	(1999)	9788498003178
Harry Potter	J.K. Rowling	(1998)	9781781103685
Magician	Raymond Feist	(1982)	9780007466863
Mistborn	Brandon Sanderson	(2006)	9788374805537
Prince of Thorns	Mark Lawrence	(2011)	9786067192681
Storm Front	Jim Butcher	(2000)	9781101128657
The Black Company	Glen Cook	(1984)	9782841720743
The Black Prism	Brent Weeks	(2010)	9782352945260
The Blade Itself	Joe Abercrombie	(2006)	9781478935797
The Colour of Magic	Terry Pratchett	(1983)	9788374690973
The Gunslinger	Steven King	(1982)	9781501143519
The Lies of Locke Lamora	Scott Lynch	(2006)	9780575079755
The Name of the Wind	Patrick Rothfuss	(2007)	9782352949152
The Painted Man	Peter Brett	(2008)	9780007518616
The Way of Kings	Brandon Sanderson	(2010)	9780765326355
The Wheel of Time	Robert Jordan	(1990)	9781857230765
Way of Shadows	Brent Weeks	(2008)	9781607513513

Note:

The short E-book numbers are the catalog entry of novels obtained from Gutenberg. Novels obtained through online purchase are denoted by the longer ISBNs.

Table A3 Overall statistics for classic and modern novels in our corpus.

Classic								
Title	Fraction of defaults	Fraction of unidentified characters	Average sentence length	Average persons per sentence	Fraction of sentences with a person	Annotated sentences	Unique characters	Total character mentions
1984	0.55	0.00 [†]	18.01	1.17	0.32	316	29	2,162
A Study in Scarlet	0.83	0.50	18.99	1.17	0.18	193	34	837
Alice in Wonderland	0.26	0.56 [°]	20.99	1.23	0.79	316	17	656
Brave New World	0.35	0.17	15.87	1.06	0.25	299	51	1,809
David Copperfield	0.61	0.00 [†]	22.79	1.08	0.49	261	157	9,922
Dracula	0.93 [°]	0.00 [†]	21.96	1.00 [†]	0.06 [†]	233	72	3,369
Emma	0.43	0.10	22.38	1.38	0.81	224	78	6,946
Frankenstein	0.86	0.22	25.80	1.19	0.17	300	29	658
Huckleberry Finn	0.59	0.14	23.46	1.20	0.40	215	82	1,749
Jekyll and Hyde	0.67	0.29	26.19	1.17	0.34	120 [†]	13 [†]	523 [†]
Moby Dick	0.88	0.38	25.24	1.10	0.10	442	135	2,454
Oliver Twist	0.36	0.33	21.64	1.23	0.68	303	69	4,495
Pride and Prejudice	0.46	0.10	24.13	1.48	0.79	257	62	5,104
The Call of the Wild	0.49	0.50	21.67	1.31	0.61	192	28	731
The Count of Monte Cristo	0.47	0.25	21.91	1.35	0.79	197	250	13,562
The Lord of the Rings	0.47	0.48	16.30	1.20	0.46	769 [°]	134	5,268
The Three Musketeers	0.60	0.36	19.19	1.13	0.49	265	115	4,842
The Way We Live Now	0.57	0.46	18.93	1.14	0.47	341	147	13,993 [°]
Ulysses	0.57	0.33	13.35 [†]	1.15	0.41	303	651 [°]	8,510
Vanity Fair	0.24 [†]	0.44	27.27 [°]	1.54 [°]	1.05 [°]	256	359	11,503
Mean μ	0.56	0.28	21.30	1.21	0.48	290.10	125.60	4,954.65
Standard Deviation σ	0.20	0.18	3.67	0.14	0.27	131.89	150.20	4,403.32
Modern								
A Game of Thrones	0.29	0.00 [†]	14.53	1.30	0.82 [°]	283	322 [°]	15,839 [°]
Assassin's Apprentice	0.71	0.29	14.94	1.18	0.38	460	66	2,857
Elantris	0.32	0.27	14.24	1.10	0.60	367	14 [†]	226 [†]
Gardens of the Moon	0.75	0.44	12.20	1.03 [†]	0.25	304	111	4,479
Harry Potter	0.32	0.33	15.55	1.33	0.74	338	84	5,114
Magician	0.49	0.17	14.78	1.16	0.45	310	115	4,976
Mistborn	0.34	0.22	12.90	1.19	0.68	297	104	11,672
Prince of Thorns	0.54	0.00 [†]	12.33	1.14	0.38	107	79	2,282
Storm Front	0.77	0.00 [†]	14.02	1.05	0.18	211	43	2,368
The Black Company	0.56	0.64 [°]	9.73 [†]	1.07	0.26	305	42	1,908
The Black Prism	0.50	0.14	13.19	1.04	0.40	380	88	10,890
The Blade Itself	0.66	0.29	12.55	1.14	0.24	103	107	6,769
The Colour of Magic	0.55	0.50	14.21	1.12	0.42	139	34	1,454
The Gunslinger	0.78 [°]	0.25	13.43	1.11	0.17 [†]	230	35	1,159
The Lies of Locke Lamora	0.21 [†]	0.09	16.90 [°]	1.38 [°]	0.77	305	105	6,477

(Continued)

Table A3 (continued).

Classic								
Title	Fraction of defaults	Fraction of unidentified characters	Average sentence length	Average persons per sentence	Fraction of sentences with a person	Annotated sentences	Unique characters	Total character mentions
The Name of the Wind	0.45	0.10	12.98	1.14	0.45	310	137	6,405
The Painted Man	0.30	0.28	14.67	1.29	0.70	301	137	9,048
The Way of Kings	0.31	0.29	12.20	1.10	0.36	316	221	14,696
The Wheel of Time	0.40	0.21	14.96	1.31	0.59	499[◊]	188	9,426
Way of Shadows	0.32	0.13	13.53	1.32	0.56	88[†]	160	8,721
Mean μ	0.48	0.23	13.69	1.17	0.47	282.65	109.60	6,338.30
Standard Deviation σ	0.18	0.17	1.54	0.11	0.20	110.52	72.98	4,535.60
$\mu_{\text{classic}} - \mu_{\text{modern}}$	0.08	0.05	7.61	0.04	0.01	7.45	16.00	-1,383.65
Pooled σ	0.20	0.17	2.46	0.24	0.25	125	119	4,473
<i>p</i> -Value	0.21	0.39	>0.01	0.73	0.74	0.85	0.68	0.35
Significant	No	No	Yes	No	No	No	No	No

Note:

The highest scores in each column are highlighted with a $\diamond$, and the lowest scores with a $\dagger$. The highest and lowest performing books for each class, in terms of F_1 -score found in Tables A3 and A4, are marked with a grey fill. Boldface indicate the highest and lowest scores in each column.

Table A4 Results of the complete BookNLP pipeline: Named entity recognition (Stanford NER), Character name clustering (e.g. ‘Tom’, ‘Tom Sawyer’, ‘Mr. Sawyer’, ‘Thomas Sawyer’ → TOM_SAWYER) and Pronominal coreference resolution.

Classic				Modern			
Title	Precision	Recall	F_1 -score	Title	Precision	Recall	F_1 -score
1984	77.33	72.87	75.03	A Game of Thrones	51.40	45.88	48.49
A Study in Scarlet [◊]	40.00	37.22	38.56	Assassin’s Apprentice [◊]	37.00	34.89	35.91
Alice in Wonderland	54.93	48.36	51.43	Elantris	72.33	73.75	73.03
Brave New World	55.00	53.57	54.28	Gardens of the Moon	12.67	14.00	13.30
David Copperfield [◊]	38.52	37.82	38.16	Harry Potter	79.17[◊]	77.78[◊]	78.47[◊]
Dracula [◊]	36.67	40.00	38.26	Magician	35.42	28.89	31.82
Emma	86.62[◊]	86.50[◊]	86.56[◊]	Mistborn	61.99	60.62	61.30
Frankenstein [◊]	51.16	45.35	48.08	Prince of Thorns	69.44	70.83	70.13
Huckleberry Finn	82.38	82.82	82.60	Storm Front [◊]	40.54	39.19	39.85
Jekyll and Hyde	52.86	50.00	51.39	The Black Company [◊]	6.85[†]	5.71[†]	6.23[†]
Moby Dick [◊]	60.98	57.72	59.31	The Black Prism	76.90	77.59	77.24
Oliver Twist	77.64	74.35	75.96	The Blade Itself	34.09	36.36	35.19
Pride and Prejudice	73.55	72.22	72.88	The Colour of Magic	30.77	27.56	29.08
The Call of the Wild	30.00	25.19	27.38	The Gunslinger	77.84	75.89	76.85
The Count of Monte Cristo	40.72	35.80	38.10	The Lies of Locke Lamora	62.77	59.16	60.91
The Fellowship of the Ring	63.23	60.61	61.90	The Name of the Wind	61.38	58.67	60.00
The Three Musketeers	13.91[†]	12.17[†]	12.99[†]	The Painted Man	60.16	57.83	58.97
The Way We Live Now	66.07	66.79	66.43	The Way of Kings	65.87	64.42	65.14
Ulysses	66.67	66.98	66.82	The Wheel of Time	29.60	24.33	26.70
Vanity Fair	72.57	68.63	70.54	Way of Shadows	54.05	45.95	49.67

Table A4 (continued).

Classic				Modern			
Title	Precision	Recall	F_1 -score	Title	Precision	Recall	F_1 -score
Mean μ	57.04	54.75	55.83	Mean μ	51.01	48.96	49.91
Standard Deviation σ	19.28	19.68	19.47	Standard Deviation σ	21.49	21.95	21.72

Note:

The highest scores in each column are highlighted with a $\diamond$, and the lowest scores with a $\dagger$. Novels written in 1st person are marked with a $\odot$. Boldface indicate the highest and lowest scores in each column.

Table A5 Social network measures for classic and modern novels.

Classic										
Title	Nodes	Edges	Average degree	Average weighted degree	Network diameter	Graph density	Modularity	Connected components	Average clustering coefficient	Average path length
1984	26	43	3.30	16.84	4	0.13	0.23	3	0.5	2.06
A Study in Scarlet	24	41	3.41	7.25	5	0.14	0.42	2	0.63	2.37
Alice in Wonderland	12	10[†]	1.66[†]	3.83[†]	3	0.15	0.15	2	0[†]	1.93
Brave New World	39	65	3.33	9.79	6	0.09	0.34	2	0.68	2.53
David Copperfield	142	499	7.03	23.11	6	0.05	0.49	2	0.57	2.69
Dracula	55	124	4.51	18.29	6	0.08	0.12[†]	4	0.52	2.53
Emma	72	403	11.19	57.53[◊]	4	0.16	0.14	1	0.67	2.16
Frankenstein	20	38	3.80	10.60	5	0.20	0.51	2	0.75	2.41
Huckleberry Finn	62	121	3.90	8.42	7	0.06	0.52[◊]	4	0.60	3.30
Jekyll and Hyde	10[†]	21	4.20	14.60	2[†]	0.47[◊]	0.12	1	0.81[◊]	1.53[†]
Moby Dick	90	169	3.76	7.38	8	0.04	0.44	8	0.59	3.33[◊]
Oliver Twist	62	191	6.16	22.32	4	0.10	0.32	2	0.75	2.26
Pride and Prejudice	62	373	12.03	57.10	4	0.20	0.16	1	0.73	1.96
The Call of the Wild	23	44	3.83	10.00	6	0.17	0.46	1	0.62	2.46
The Count of Monte Cristo	228	799	7.01	24.05	7	0.03	0.40	3	0.56	2.88
The Fellowship of the Ring	105	260	4.95	11.51	6	0.05	0.29	2	0.63	2.73
The Three Musketeers	96	279	5.81	15.33	5	0.06	0.32	1	0.55	2.56
The Way We Live Now	135	630	9.33	39.17	5	0.07	0.36	3	0.69	2.43
Ulysses	522[◊]	4,116[◊]	15.77[◊]	18.59	9[◊]	0.03	0.45	10[◊]	0.60	3.02
Vanity Fair	342	1,349	7.89	22.73	7	0.02[†]	0.37	1	0.63	2.72
Mean μ	106	479	6.14	20	5.45	0.12	0.33	2.75	0.60	2.49
Standard Deviation σ	126.94	916.66	3.56	14.99	1.70	0.10	0.14	2.39	0.17	0.44
Modern										
A Game of Thrones	314[◊]	1,648[◊]	10.50[◊]	22.46	6	0.03	0.48	1	0.54	2.81
Assassin's Apprentice	55	110	4.00	9.09	6	0.07	0.34	2	0.49	2.65
Elantris	106	493	9.30	43.25[◊]	5	0.09	0.36	1	0.67	2.22[†]
Gardens of the Moon	88	257	5.84	10.84	8	0.07	0.42	1	0.48	2.93
Harry Potter	67	198	5.9	19.37	5	0.09	0.15	1	0.68	2.23
Magician	84	209	4.98	10.76	6	0.06	0.43	2	0.58	2.83

(Continued)

Table A5 (continued).

Classic										
Title	Nodes	Edges	Average degree	Average weighted degree	Network diameter	Graph density	Modularity	Connected components	Average clustering coefficient	Average path length
Mistborn	89	255	5.73	33.89	6	0.07	0.04 [†]	3	0.62	2.37
Prince of Thorns	59	111	3.76	6.98	6	0.07	0.37	2	0.42 [†]	2.83
Storm Front	33	85	5.15	10.97	4 [†]	0.16 [°]	0.31	1	0.64	2.26
The Black Company	30	45	3.00 [†]	6.13 [†]	6	0.10	0.20	3	0.561	2.43
The Black Prism	84	239	5.69	30.74	5	0.07	0.22	1	0.75 [°]	2.27
The Blade Itself	96	259	5.40	14.23	5	0.06	0.51	3	0.51	2.65
The Colour of Magic	27 [†]	43 [†]	3.19	7.93	6	0.12	0.38	1	0.50	2.67
The Gunslinger	31	69	4.45	8.52	7	0.15	0.41	1	0.43	2.87
The Lies of Locke Lamora	101	261	5.17	22.24	5	0.05	0.18	4	0.64	2.46
The Name of the Wind	109	197	3.62	8.99	9 [°]	0.03	0.67 [°]	5	0.46	4.06 [°]
The Painted Man	132	444	6.73	23.15	7	0.05	0.53	1	0.63	2.70
The Way of Kings	172	448	5.21	20.79	6	0.03 [†]	0.57	9 [°]	0.55	2.91
The Wheel of Time	167	545	6.53	16.66	7	0.04	0.35	3	0.55	2.84
Way of Shadows	145	441	6.08	22.14	6	0.04	0.46	4	0.61	2.71
Mean μ	99	317	5.50	17	6.05	0.07	0.36	2.45	0.56	2.68
Standard Deviation σ	66.37	348.92	1.85	10.05	1.15	0.04	0.15	1.99	0.09	0.4
$\mu_{\text{classic}} - \mu_{\text{modern}}$	7	162	0.64	3	-0.60	0.05	-0.03	0.30	0.04	-0.19
Pooled σ	101	695	2.83	12.83	1.45	0.08	0.15	2.18	0.13	0.43
p -Value	0.83	0.47	0.49	0.55	0.20	0.09	0.42	0.67	0.37	0.17
Significant	No	No	No	No	No	No	No	No	No	No

Note:

The highest scores in each column are highlighted with a $\diamond$, and the lowest scores with a $\dagger$. The highest and lowest performing books for each class, in terms of F_1 -score found in Tables 3 and 4, are marked with a grey fill. Boldface indicate the highest and lowest scores in each column.

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Competing Interests

The authors declare that they have no competing interests.

Author Contributions

- Niels Dekker conceived and designed the experiments, performed the experiments, analysed the data, contributed reagents/materials/analysis tools, prepared figures and/or tables, performed the computation work, authored or reviewed drafts of the paper, approved the final draft.
- Tobias Kuhn contributed reagents/materials/analysis tools, authored or reviewed drafts of the paper, approved the final draft.

- Marieke van Erp conceived and designed the experiments, contributed reagents/materials/analysis tools, authored or reviewed drafts of the paper, approved the final draft.

Data Availability

The following information was supplied regarding data availability:

Code and data are available at GitHub: <https://github.com/Niels-Dekker/Out-with-the-Old-and-in-with-the-Novel>.

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